SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Nighil Adhithya.R *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	1	2	2
Explanation	25%	1	2	2	1
Application	20%	2	2	1	1
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	2
Clarity	5%	0	0	0	0

Total Marks 28 **/40**

Signature of the Faculty

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Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of

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3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

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Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organizatio n	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professiona l presentatio n	Understanda ble with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

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Computer Networks short answers Test (CO1 - AT1)

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6) Given:

Original file size = 900 MB

Compression = 40%

Encryption overhead = 5% of compressed size

$$\text{Compressed size} = 900 \times \frac{100-40}{100}$$

$$= 900 \times 0.6 = 540 \text{ MB}$$

$$\text{overhead} = 540 \times \frac{5}{100}$$

$$= 27 \text{ MB}$$

$$\text{Encrypted size} = 540 + 27 = 567 \text{ MB}$$

$$\%\ \text{Reduction} = \frac{900-567}{900} \times 100$$

$$= \frac{333}{900} \times 100$$

$$= 37\%$$

The presentation layer improves the transition transmittion efficiency in following ways:

- compression.
- Encryption.
- Data formatting.

7) Given data:

Total data = 3GB

Effective throughput = 120Mbps

Request size = 3MB

Assuming 1GB = 1000MB

Total data = 3000MB

No. of requests = Total data / Request size = 3000/3 = 1000 requests

8 × 3000 MB = 24000 Mb

$$\text{Time} = \frac{\text{Total data in Mb}}{\text{Throughput in Mbps}} = \frac{24000}{120} = 200 \text{ seconds}$$

The application layer provides network services directly to user applications. These are the benefits of it:

- File transfer
- Error handling
- Authentication
- Data management
- Reliable transfer.

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Given data:

Component	Delay
Application layer	4 ms
Transport layer	3 ms
Network layer	5 ms
Data link layer	2 ms
Physical layer	1 ms
propogation delay	18 ms
Transmission delay	12 ms

Total delay = Processing delays + Propogation delay + Transmission delay

Sum of processing delays = 4+3+5+2+1 = 15 ms

Total delay = 15 + 18 + 12 = 45 ms

Application layer : provides network services to user applications.

Transport layer : Handles segmentation, port addressing & Reliable delivery when using TCP.

Network layer : Determines the routes and forwards packets via IP addressing.

Datalink layer : Creates frames & handles local network addressing

physical layer : Transmits bits with a medium.

propogation delay : Time taken for the signal to travel through the medium.

Transmission delay : Time required to place all packet bits into one medium.

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9) Given data:

Useful data : 80MB.

Frame size : 1600 bytes.

Protocol overhead per frame } : 80 Bytes.

Useful data per frame } : 1600 - 80 = 1520 bytes.

Assume : 1Mb = 10^6 Bytes.

Useful data = 8×10^6 Bytes.

$$\text{No. of frames} = \frac{8 \times 10^6}{1520} = 52631.58 \approx 52,632 \text{ frames.}$$

Total overhead = No. of frame × overhead per frame.

= 52,632 × 80

= 4,210,560 Bytes ≈ 4.21MB

$$\% \text{ Efficiency} = \frac{\text{Useful data}}{\text{Total transmitted data}} \times 100$$

$$= \frac{1520}{1600} \times 100$$

= 95%

Effect of protocol overhead on Network performance

- Reduces efficiency.
- Increased bandwidth usage.
- Can Increase transmission time.
- Supports communication.

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